

Buck-Boost converter and CUK converter

Q1. Design and derive the expression for L value for a buck-boost converter for the following specifications: $V_s = 40V$, $V_o = -60V$, $R_{Load} = 20 \Omega$, $I_{LMAX} = 10.7A$ and $I_{LMIN} = 4.3A$ with a switching period of 200 micro sec. For a peak-to-peak output voltage ripple is 2V, obtain the value of C with the obtained safety factor of continuous conduction mode so that the output voltage ripple is less than 2V. Also calculate I_{CRMS} .

Q2. Consider a buck-boost regulator that supplies 75W at $I_o = 5A$ from 37V DC source. Let $T = 130\mu\text{sec}$ and $L = 250\mu\text{H}$. Determine (a) duty ratio, D, (b) I_{LMAX} and I_{LMIN} (c) average input current (d) average diode current and transistor current (e) RMS value of capacitor current.

Q3. The input voltage of a Cúk converter is $V_s = 12V$. The duty cycle is 0.25 and the switching frequency is 25 kHz. The filter inductance is $L_2 = 150 \mu\text{H}$ and filter capacitance is $C_2 = 220 \mu\text{F}$. The energy transfer capacitance is $C_1 = 200 \mu\text{F}$ and input inductance $L_1 = 180 \mu\text{H}$. The average load current is $I_o = 1.25 A$. Assuming continuous current mode in both the inductors, determine (a) the average output voltage; (b) the average input current I_s ; (c) the peak-to-peak ripple current of inductor L_1 ; (d) the peak to- peak ripple voltage of capacitor C_1 ; (e) the peak-to-peak ripple current of inductor L_2 ; (f) the peak-to-peak ripple voltage of capacitor C_2 .

Q4. A Cúk converter has an input of 12V and is to have an output of -18V supplying a 40W load. Select the duty ratio for a switching frequency of 50Hz, assuming continuous conduction in both inductors. Design the inductors such that the change in the corresponding inductor currents is no more than 10% of the average inductor currents. Given that the output ripple voltage is no more than 1% and the ripple voltage across C_c is no more than 5%.
